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Feature Extraction

Question

Explain the 3 main concepts of feature extraction based on supervised learning, auto-encoders, and self-supervised learning.

Give examples of using these features.

Introduction

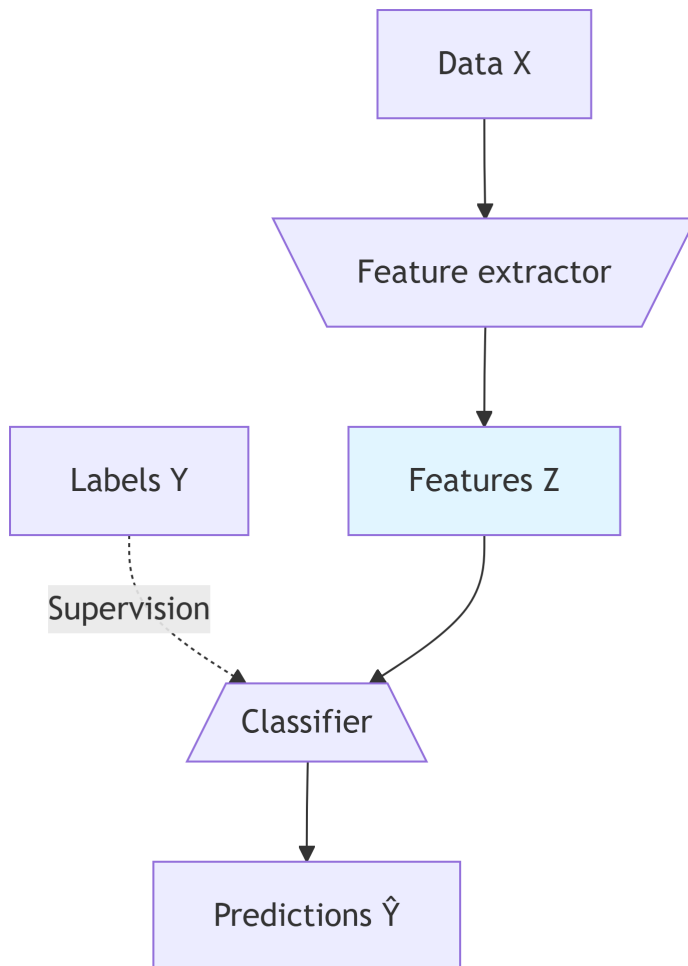
Feature extraction transforms raw data into compact, informative representations. Three main approaches exist:

1. **Supervised learning:** uses labeled data
2. **Auto-encoders:** reconstructs data through a bottleneck
3. **Self-supervised learning:** creates its own learning tasks

Each method seeks to optimize different aspects of information, related to entropy, KL divergence, and mutual information.

1. Supervised Learning

Principle and Link to Information Theory



Objective: Maximize the **mutual information** between features Z and labels Y .

Intuition: We seek to extract features that maximally reduce uncertainty about the labels. If Z contains much information about Y , then knowing Z strongly reduces the entropy of Y .

Application Domains

Extraction (Feature Extraction)

- **Example:** ResNet for images - convolutional layers hierarchically extract patterns (edges → textures → object parts)
- **Theory link:** Features optimized for discrimination

Mining (Exploration)

- **Example:** Sentiment analysis in customer reviews
- **Usage:** Discover recurring themes in negative comments
- **Approach:** Supervised classification then analysis of important features

Recognition & Classification

- **Example:** Facial recognition for authentication
- **Principle:** 2-ary or M-ary classification - decide between “person A” vs “person B” vs “others”
- **Performance:** High accuracy when plenty of labeled data

Search & Identification

- **Example:** Image-based product search engine
- **Mechanism:** Extracted features compared by cosine distance
- **Advantage:** High precision thanks to discriminative learning

Transmission & Compression

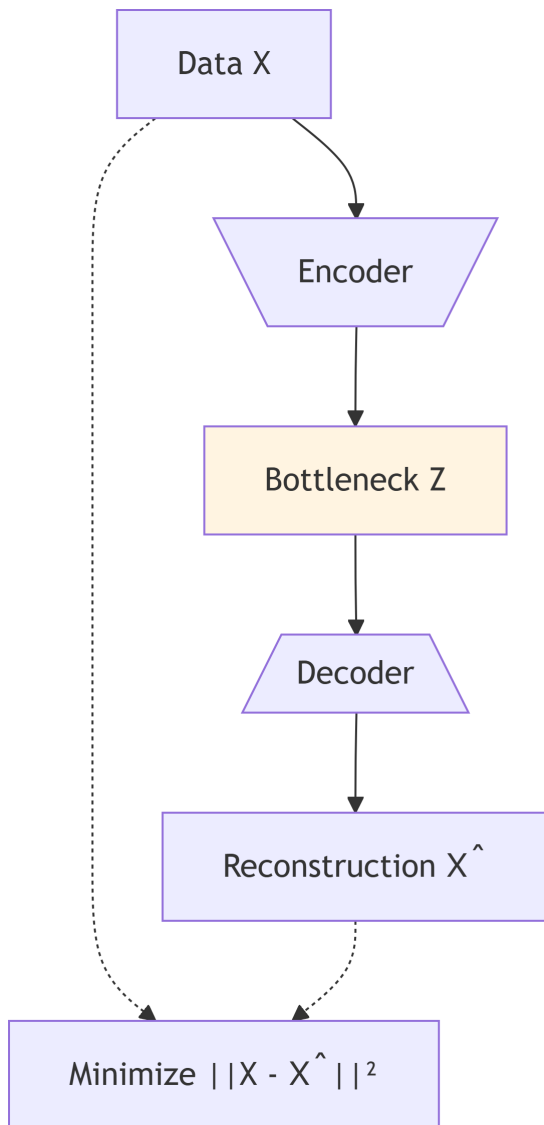
- **Example:** Intelligent compression - keep only discriminative features
- **Approach:** Learning compressed representations for classification

Multimedia Security

- **Example:** Deepfake detection
- **Principle:** Binary classification (real/fake) based on learned features
- **Features:** Capture generation artifacts

2. Auto-Encoders

Principle and Link to Information Theory



Objective: Minimize the **KL divergence** between the data distribution and their reconstruction.

Intuition: We seek a compressed representation Z that preserves essential information. It's a compromise between compression (small Z) and fidelity ($\hat{X}$ close to X).

Application Domains

Extraction (Feature Extraction)

- **Example:** Non-linear dimensionality reduction
- **Usage:** Projection of complex data to 2D/3D for visualization
- **Advantage** vs PCA: Captures non-linear relationships

Mining (Exploration)

- **Example:** Community discovery in social networks
- **Approach:** Auto-encoder on adjacency matrix
- **Result:** Clusters corresponding to natural communities

Recognition & Classification

- **Example:** Medical anomaly detection
- **Principle:** Only normal images reconstruct well
- **Decision:** High reconstruction error $\rightarrow$ anomaly

Search & Identification

- **Example:** Music recommendation system
- **Mechanism:** Similarity search in latent space Z
- **Advantage:** Captures deep structural similarity

Transmission & Compression

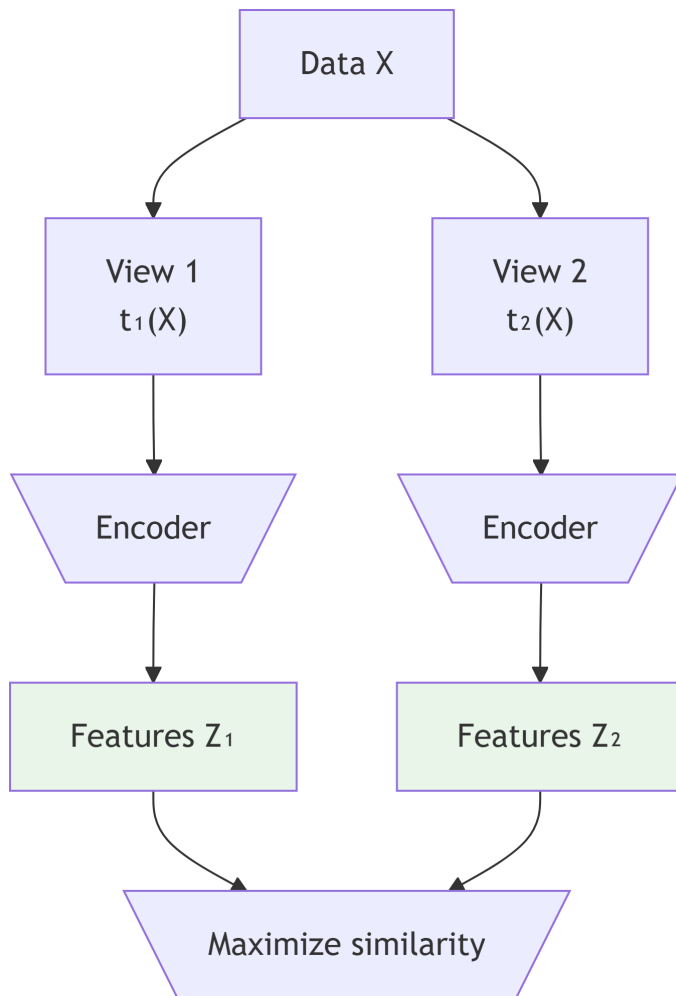
- **Example:** Neural video codecs (e.g., BPG, VVC)
- **Principle:** Lossy compression optimized for human perception
- **Trade-off:** Rate-distortion - size/quality compromise

Multimedia Security

- **Example:** Robust watermarking
- **Approach:** Encode watermark in bottleneck Z
- **Robustness:** Survives transformations as long as reconstruction is faithful

3. Self-Supervised Learning

Principle and Link to Information Theory



Objective: Maximize the **mutual information** between different views of the same data.

Intuition: Two transformations of the same image should have similar representations. Thus we learn features invariant to transformations but discriminative between different data.

Application Domains

Extraction (Feature Extraction)

- **Example:** BERT models in NLP

- **Principle:** Masked word prediction → rich contextual embeddings
- **Usage:** Base for many NLP tasks

Mining (Exploration)

- **Example:** Category discovery in unlabeled dataset
- **Approach:** Contrastive pre-training then clustering
- **Result:** Emerging categories without human supervision

Recognition & Classification

- **Example:** Medical diagnosis with few labeled data
- **Protocol:** 1) Self-supervised pre-training on all available medical images 2) Fine-tuning on small labeled dataset
- **Advantage:** Drastic reduction in annotation needs

Search & Identification

- **Example:** Cross-modal image-text search (CLIP)
- **Capability:** Find images from textual descriptions
- **Usage:** Advanced visual search engines

Transmission & Compression

- **Example:** Semantic compression
- **Idea:** Transmit only essential semantic information
- **Approach:** Self-supervised pre-training to identify what is semantically important

Multimedia Security

- **Example:** AI-generated content detection
- **Principle:** Generative models leave subtle fingerprints
- **Detection:** Self-supervised features capture these patterns

Summary Table of Theoretical Links

Method	What to optimize?	Link with information theory	Intuitive analogy
Super-vised	Correct label prediction	Maximize $I(Z;Y)$ - mutual information features/labels	“Learn to distinguish”
Auto-encoders	Faithful reconstruction	Minimize $D_{KL}(X\ X)$ - divergence between original and reconstruction	“Learn to remember the essential”
Self-supervised	Similarity between augmented views	Maximize $I(Z;Z)$ - mutual information between views of same data	“Learn to recognize despite transformations”

Integrated Concrete Examples

Case 1: Video streaming platform

Problem: Recommend movies to users

Hybrid solution:

1. **Self-supervised:** Pre-training on all videos (learn to recognize scenes, actions)
2. **Supervised:** Fine-tuning on user ratings
3. **Auto-encoders:** Compression of embeddings for fast search

Applications:

- **Mining:** Discover emerging genres (e.g., “Christmas romantic comedies”)
- **Search:** “Find movies similar to Inception”
- **Compression:** Adaptive streaming based on bandwidth

Case 2: Online bank

Problem: Detect fraudulent transactions

Approach:

1. **Auto-encoder:** Normal model on legitimate transactions
2. **Detection:** Transactions with high reconstruction error = potential fraud
3. **Supervised:** Human validation → label creation → more precise supervised model

Security: Real-time anomaly detection

Conclusion: Which Method to Choose?

Need	Recommended Method	Why
Plenty of labeled data	Supervised	Maximum performance on specific task
Compression, anomalies	Auto-encoders	Optimized for faithful reconstruction
Few labels, lots of raw data	Self-supervised	Exploits natural data structure
Complete system	Combination of all three	Self-supervised pre-training → supervised fine-tuning → auto-encoder compression

Current trend: Self-supervised learning is becoming dominant as it exploits the abundance of unlabeled data, followed by supervised fine-tuning to adapt to specific tasks, with auto-encoders for operational efficiency.

Comparative Table and Synthesis

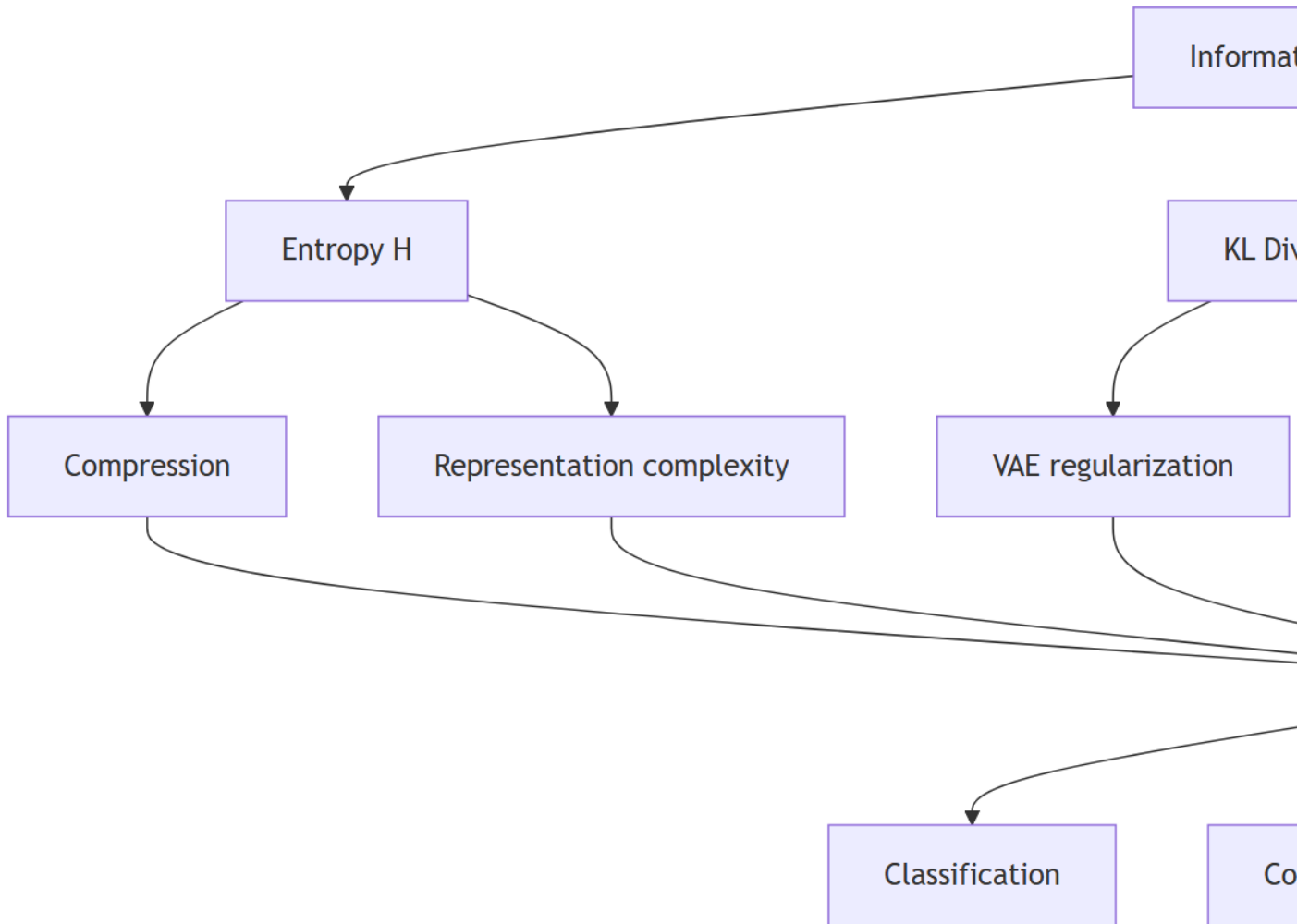
Aspect	Supervised Learning	Auto-encoders	Self-Supervised Learning
Data	Labeled	Unlabeled	Unlabeled
Objective	Predict Y	Reconstruct X	Learn general representations
Link with info theory	$\text{Max } I(Z; Y)$	$\text{Min } D_{\text{KL}}, \text{ bottleneck}$	$\text{Max } I(Z_1; Z_2), \text{ InfoNCE}$
Characteristics	Task-specific	Compact, reconstructible	Generalizable, transferable

Aspect	Supervised Learning	Auto-encoders	Self-Supervised Learning
Typical applications	Classification, regression	Compression, anomalies	Pre-training, transfer learning

Relations with Entropy, DKL, and Mutual Information

1. **Entropy** $H(\cdot)$: Measures uncertainty/information content
 - In all cases, we seek representations with appropriate entropy (sufficiently informative but not too redundant)
2. **KL Divergence** $D_{\text{KL}}(p \parallel q)$: Measures discrepancy between distributions
 - VAE: $D_{\text{KL}}(q(z|x) \parallel p(z))$ regularizes latent space
 - Domain adaptation: Distribution alignment
 - Self-supervised: Minimize DKL between augmented views
3. **Mutual Information** $I(\cdot; \cdot)$: Measures dependence
 - Supervised: Maximize $I(Z; Y)$
 - Self-supervised: Maximize $I(Z_1; Z_2)$
 - Autoencoders: Preservation of $I(X; Z)$ under compression constraint

Synthesis Diagram of Relations



Conclusion

The three feature extraction paradigms offer complementary approaches with deep links to information theory:

1. **Supervised learning** directly maximizes mutual information with labels, producing highly discriminative features for a specific task.
2. **Auto-encoders** implement an information bottleneck with KL divergence regularization, learning compact representations that preserve essential information.

3. **Self-supervised learning** maximizes mutual information between different views of the data, learning general and transferable representations.

These theoretical links are not merely academic: they guide the development of more efficient architectures and explain why some methods work better than others in specific contexts. Understanding these connections allows practitioners to choose the appropriate approach for their problem and innovate by combining the strengths of each paradigm.